Networks and Distributed Programming

Introduction to Computer Networks

Lecture 1



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II



DIETI

UNINA

Computer networks

Computer network

- A collection of interconnected, autonomous computing devices
- Interconnected computers can exchange information
- Information is exchanged to perform tasks and provide services

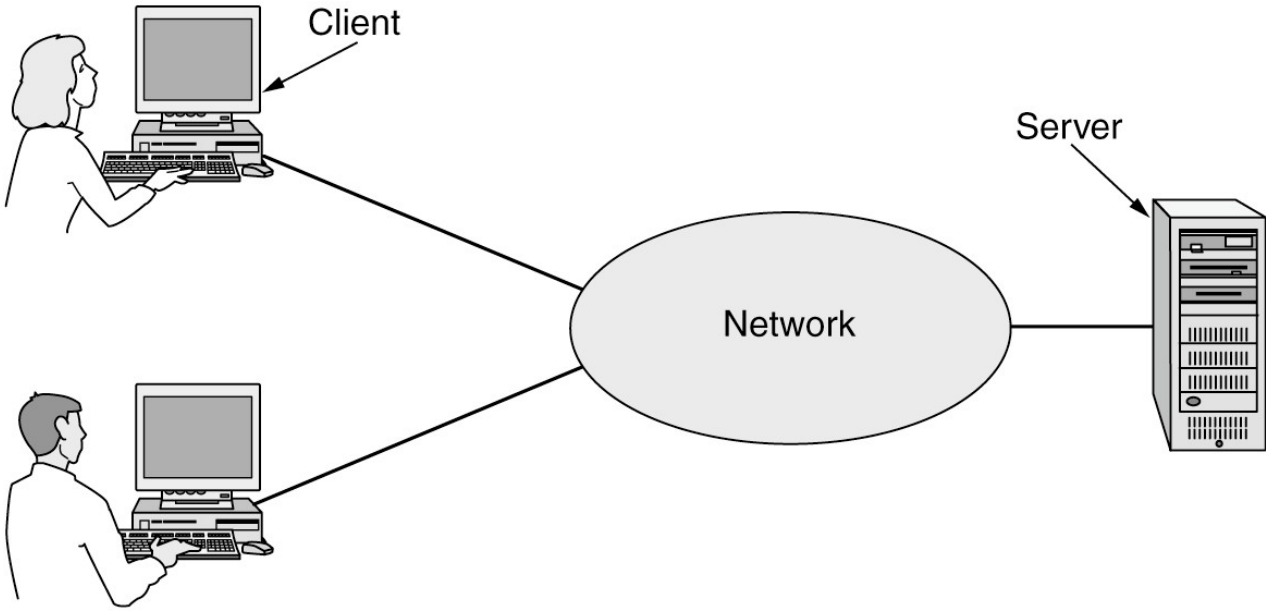
> ***Computer networking*: *the process of connecting computer together so that they can share information.***
>
> [Cambridge Dictionary]

What networks enable

- Access to remote information and services
- Communication and collaboration
- Distributed computation and cloud services
- Media, commerce, and online services
- Internet of Things and cyber-physical systems

Access to Information

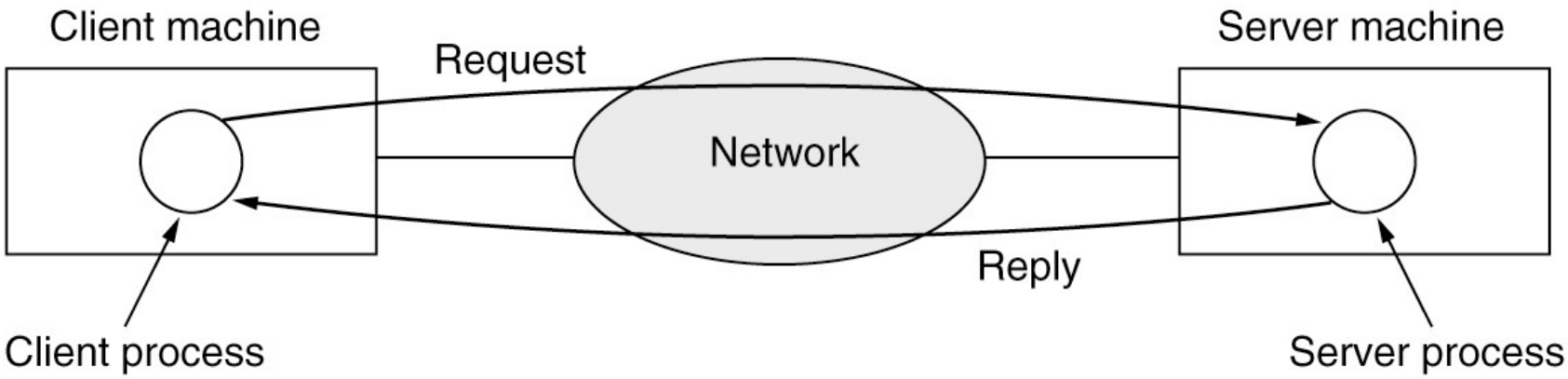
Client-server



In the client-server model, a client explicitly requests information from a server that hosts that information.

Access to information

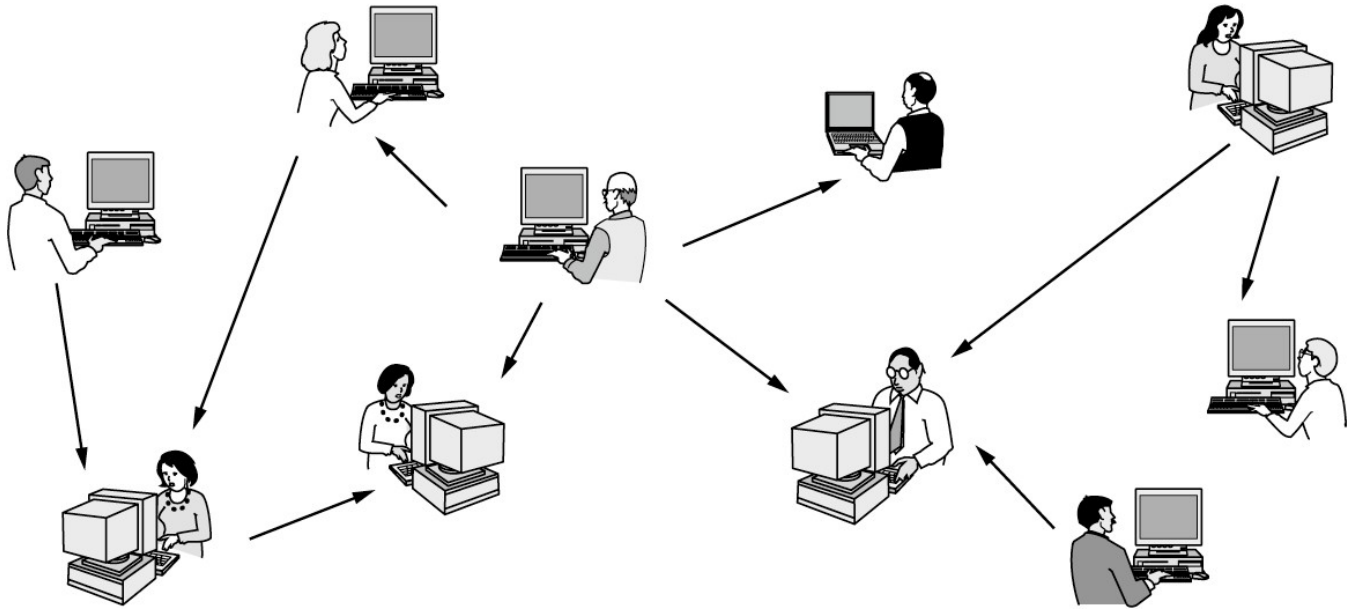
Client-server



Communication takes the form of the client process sending a message over the network to the server process. The client process then waits for a reply message.

Access to information

Peer-to-peer



In a peer-to-peer system, there are no fixed clients and servers.

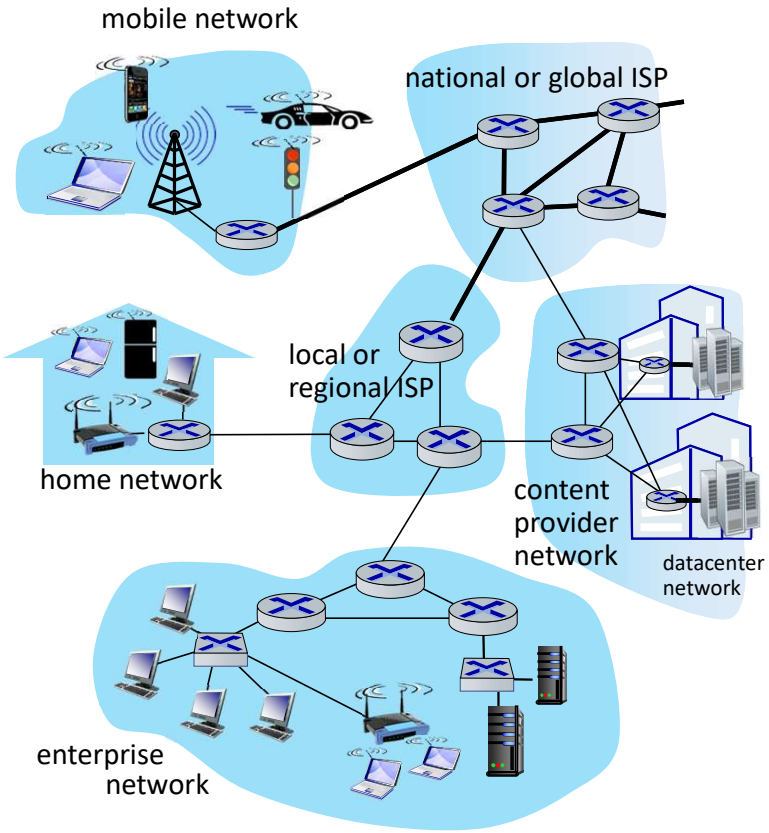
Introduction
Computer Networks and Internet

Internet

- A global **network of interconnected networks**
- It connects heterogeneous networks: home, mobile, enterprise, data center networks …
- It provides the communication infrastructure used by a huge variety of services.

Not all networks are part of the Internet

- Networks can be local, private, and disconnected from the Internet



Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.



Internet is a network of networks. It spans countries and continents, interconnecting networks over very large geographical areas.

The idea of such a “universal” network was envisioned by researchers in the early 1960s (e.g., J.C.R. Licklider's “Galactic Network”, 1962). ARPANET, launched by ARPA in 1969, was a major step towards this vision.

Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.

- Brief history of Internet:
  - The Advanced Research Projects Agency (ARPA) of the U.S. Department of Defense awarded contracts for the development of the **ARPANET** project (1969).
  - **Goal:** connect research centers and share expensive computing resources.
  - **Packet switching:** data are divided into packets and forwarded through intermediate nodes.
  - Each site used a dedicated **Interface Message Processor (IMP)** to connect the local host to the network and forward packets.
  - **First ARPANET message:** UCLA → SRI, October 1969 ("LO").



Initial ARPANET having 4 nodes: SRI, UCLA, UCSB, and UTAH (1969)

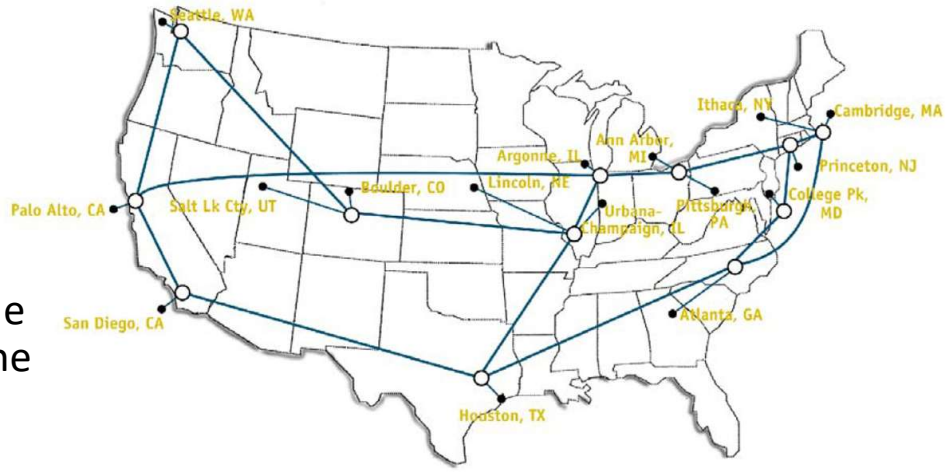
The initial ARPANET was mainly a closed network designed to include universities and research centers. Similar networks were independently created all around US and Europe.

Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.

- Brief history of Internet:
  - The Advanced Research Projects Agency (ARPA) of the U.S. Department of Defense awarded contracts for the development of the **ARPANET** project (1969).
  - R. Kahn (ARPA) and V. Cerf (Stanford) proposed a protocol architecture to **interconnect different packet-switched networks** (1974), laying the foundations of TCP/IP.
  - ARPANET officially switched from NCP to TCP/IP (1983)
  - NSFNET was launched as a TCP/IP backbone connecting NSF supercomputer centers, universities, and regional academic networks (1986).



NSFNET in 1992 (Wikipedia).

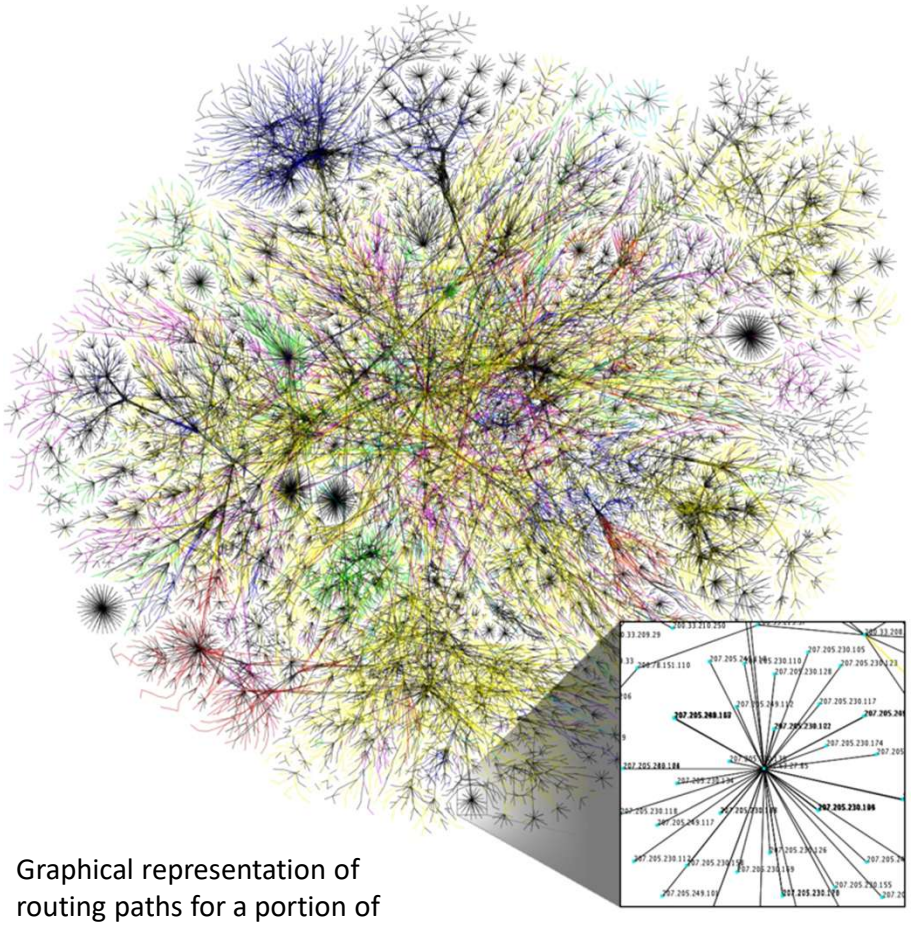
The NSFNET and the TCP/IP protocol went toward the idea of a *network of networks,* but both ARPANET and NSFNET carried **commercial restrictions**.

Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.

- Brief history of Internet:
  - The Advanced Research Projects Agency (ARPA) of the U.S. Department of Defense awarded contracts for the development of the **ARPANET** project (1969).
  - R. Kahn (ARPA) and V. Cerf (Stanford) designed the Transmission Control Protocol (TCP) and Internet Protocol (IP), two protocols of the **Internet protocol suite** (1974).
  - The National Science Foundation (NSF) awarded contracts for the **NSFNET** project, a TCP-IP based network (1986).
  - The **ARPANET decommissioned** (1990).
  - The **NSFNET decommissioned**, removing the last restrictions on the use of the Internet for commercial traffic (1995).

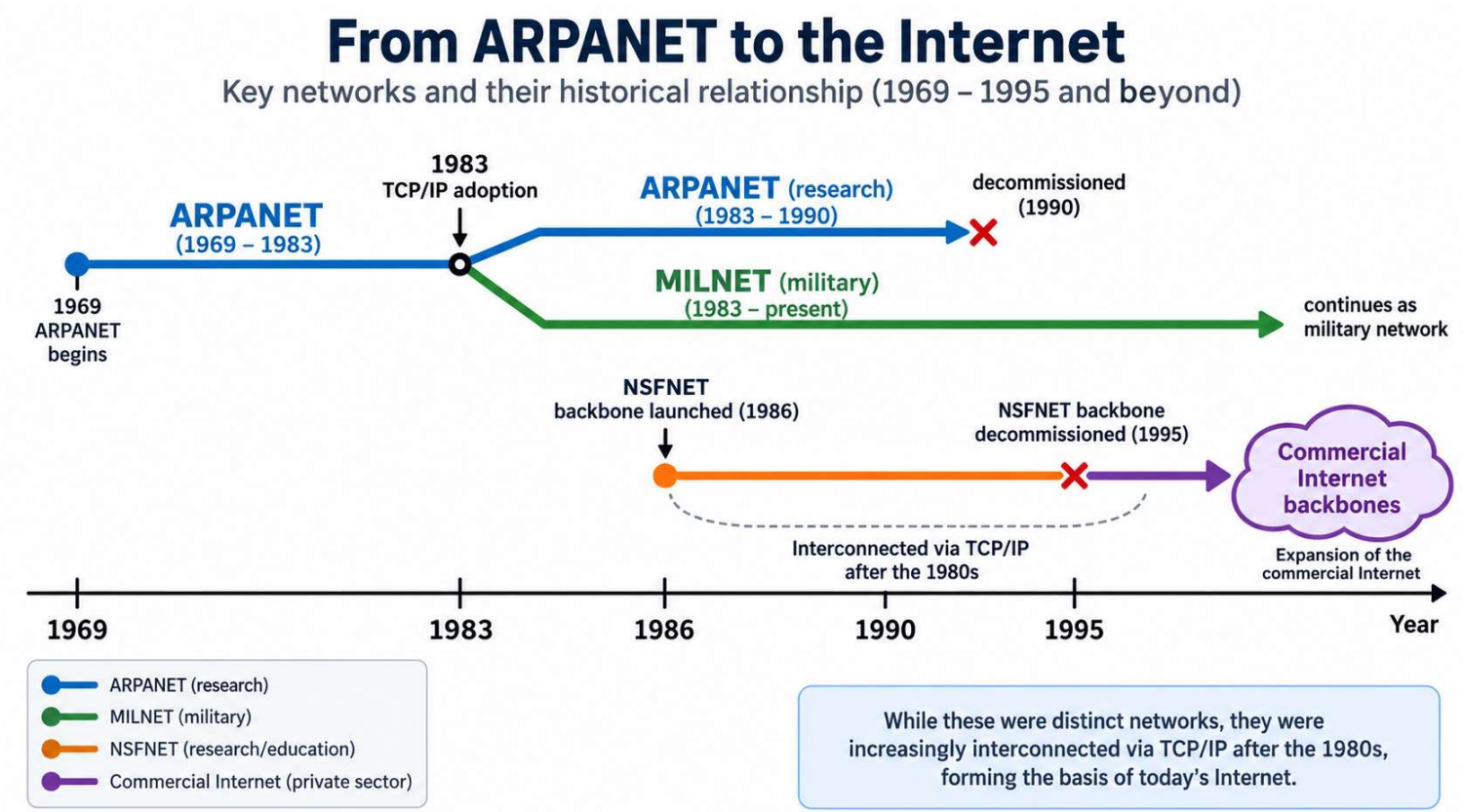


Graphical representation of routing paths for a portion of Internet (Wikipedia).

Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.



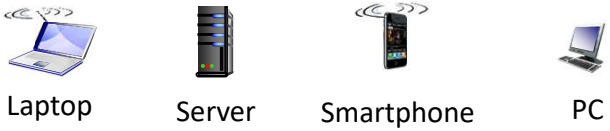
Introduction

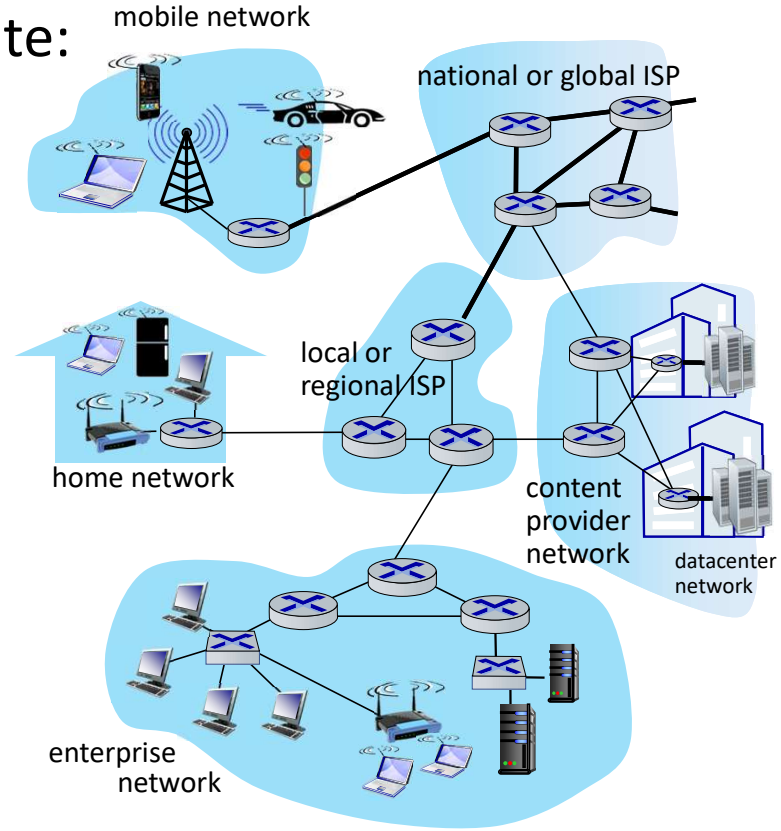
Networks Components: Devices, Links, Hosts

- Network **devices** and **links** form the **infrastructure** that allow hosts to communicate:



- **Hosts** are devices on which **applications** (or programs) runs:



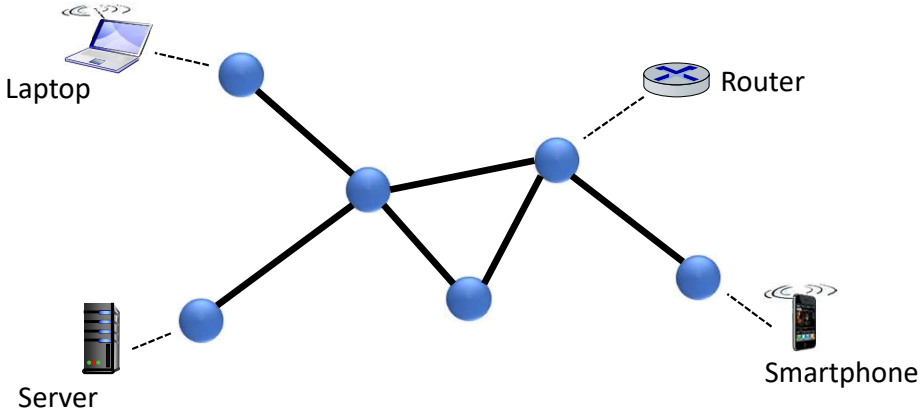


Introduction

Computer Networks and graphs

- Computer networks share terminologies with graph theory:
  - The devices connected through the network are called **nodes** while the connections between nodes are called communication **links** (or **channels**).
  - A **sequence** of nodes/links is called **path**.

- The end-points (or end-systems) of the network, which provide or use **services**, are special nodes called **hosts**.



Ideally hosts are the leaves of the network (often it is not true) while intermediate nodes are routing devices (routers, switches).

Introduction
Data Communication

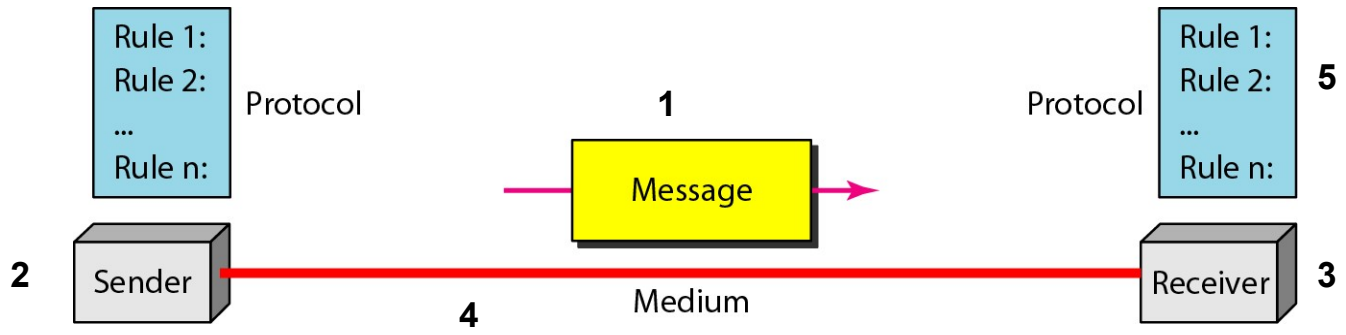
- The **goal of networking** is to share information: to allow large-scale internet functionalities as well as small-scale network functionalities we need to connect two hosts so they can **communicate data over distance**.

- **Data communication** is the process of exchanging data between two devices via some form of transmission medium (e.g., wires or wireless), it is often a **tradeoff** between:
  - **Reliability**: data is received correctly.
  - **Performance**: data is received within a reasonable amount of time.

- The **telecommunication** (i.e., communication at a distance) is the field that studies how to communicate data over distance.

Introduction
Data Communication

- Data communication main components:
  1. **Message**: contains the **data** to be transmitted.
  2. **Sender**: the entity which is sending the message.
  3. **Receiver**: the entity which is supposed to receive the message.
  4. **Medium**: the channel between **sender** and **receiver** where the **message** travels.
  5. **Protocol**: a set of communication rules, known to **sender** and **receiver**.



Introduction

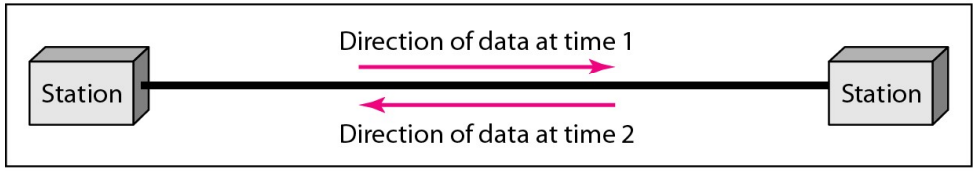
Data Representation and Flow

- The **data** we want to exchange can be represented in different forms.
  - Text, numbers, images, audio, video, etc.

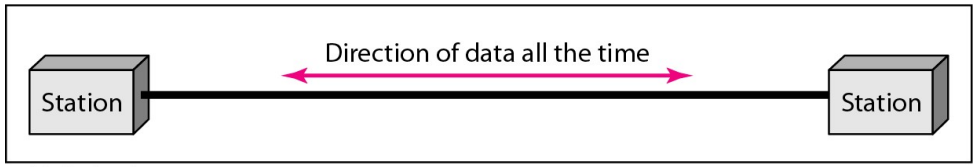
- Depending on the type and the purpose of the communication, the **data flow** can be:
  - **Simplex**: monodirectional.
  - **Half-duplex**: bidirectional (taking turns).
  - **Full-duplex**: bidirectional (simultaneous).



a. Simplex



b. Half-duplex



c. Full-duplex

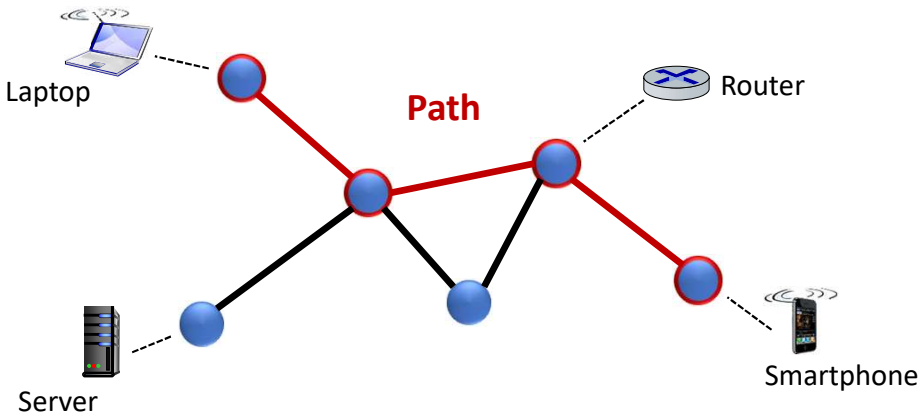
Introduction

Data Representation and Flow

- **Transmission rate:** the nominal rate at which bits are transmitted over a link, measured in bit/s.

- **Link/bandwidth capacity:** the **maximum data rate** that a link can support. For an end-to-end path, the maximum rate is limited by its bottleneck link.

- **Throughput:** the actual data rate achieved by a communication over a given time interval, typically measured in bit/s.

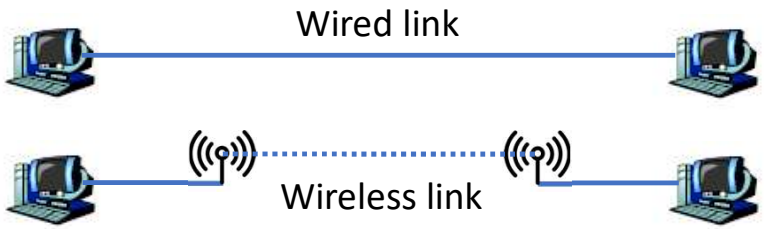


Introduction

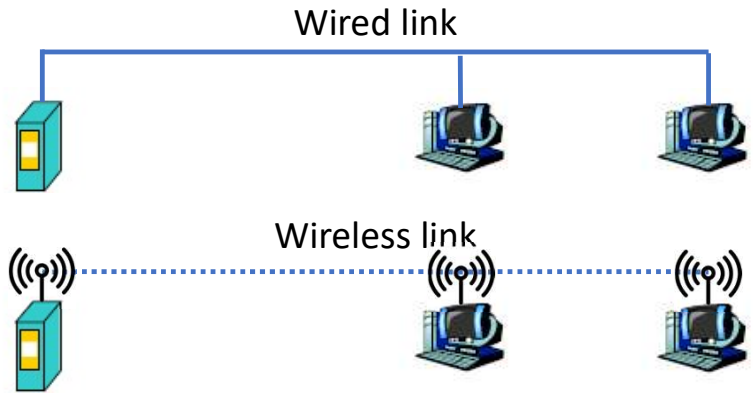
Types of Connections

- The hosts of a network can be connected in different ways.

- **Point-to-point**: a dedicated link is provided between two devices (wireless or wired).



- **Multipoint** (**shared-medium**): more than two specific devices share a single link (wireless or wired).

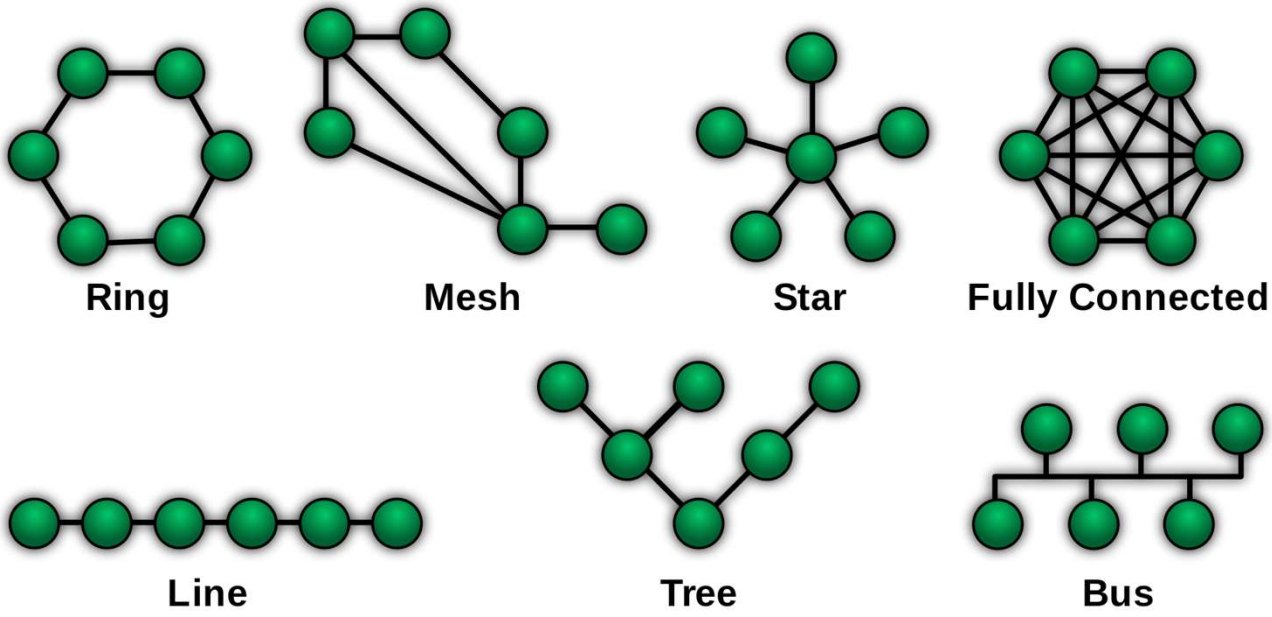


Introduction

Network Topologies

- There are different ways of connecting devices.

- The **network topology** is the arrangement of the elements (links, nodes, etc.) of a communication network.



en.wikipedia.org

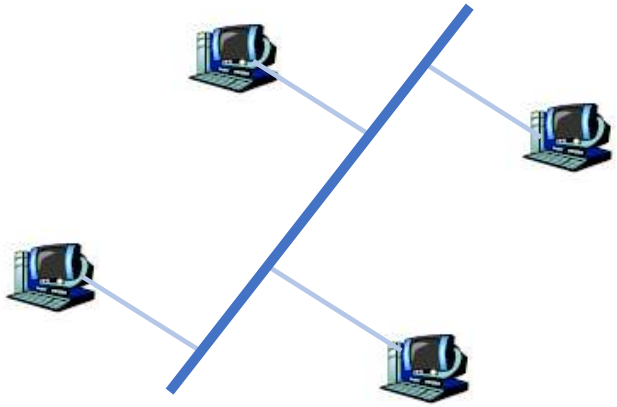
Introduction

Network Topologies: Bus

- In the **Bus** topology hosts are connected to a central backbone (bus) cable.
  - **Collisions** are possible.

- **Pros:**
  - Simple and cheap.
  - Good for small networks.

- **Cons:**
  - Single point of failure (broken bus) but sub networks may still be available.

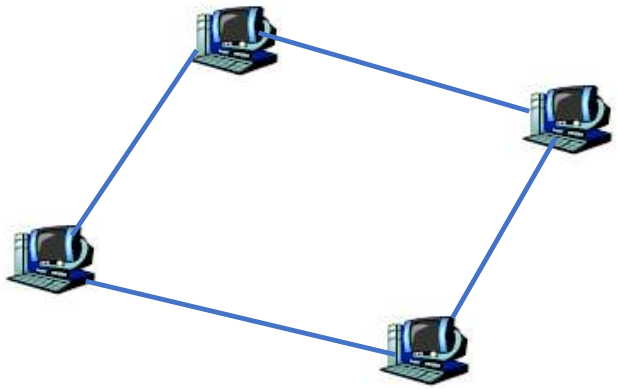


A bus topology consists of one shared backbone, with n hosts attached to it.

Introduction

Network Topologies: Ring

- In the **Ring** topology hosts are point-to-point connected to exactly two other ones. The signal is **forwarded** along the ring, from device to device, until it reaches its destination.



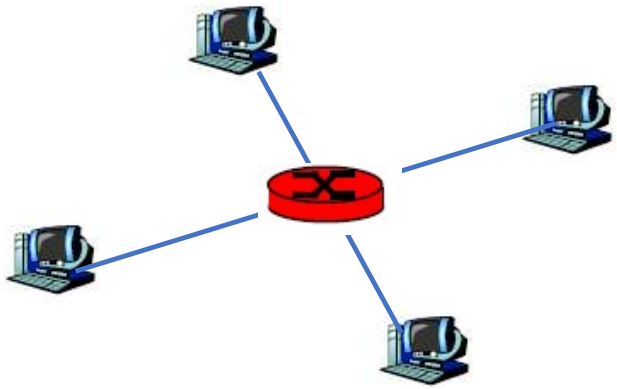
- **Pros:**
  - Simple and cheap.
  - Performs better than the bus.
- **Cons:**
  - Adding new nodes is harder.
  - Nodes malfunctioning may impair the network.

Ring networks have *n* point-to-point links.

Introduction

Network Topologies: Star

- In the **Star** topology hosts are linked to a central node (hub, switch or router), there is no direct link between hosts.
- The central node redirects messages.



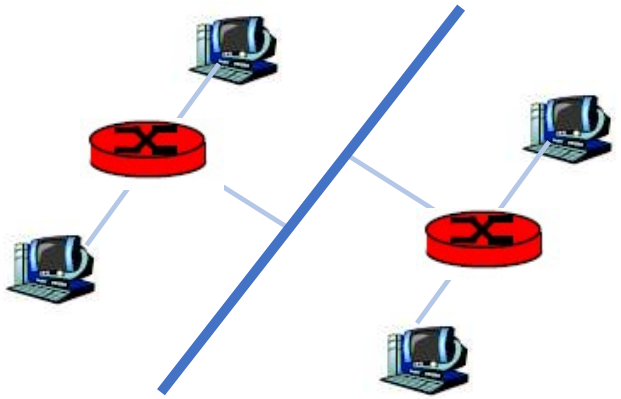
- **Pros:**
  - Less expensive, simple, robust, more scalable.
- **Cons:**
  - The controller must be reachable by all hosts.
  - Single point of failure.

Star networks have *1* central node and *n* point-to-point links

Introduction

Network Topologies: Tree

- **Tree topology:** nodes are organized hierarchically, typically by interconnecting multiple star networks.



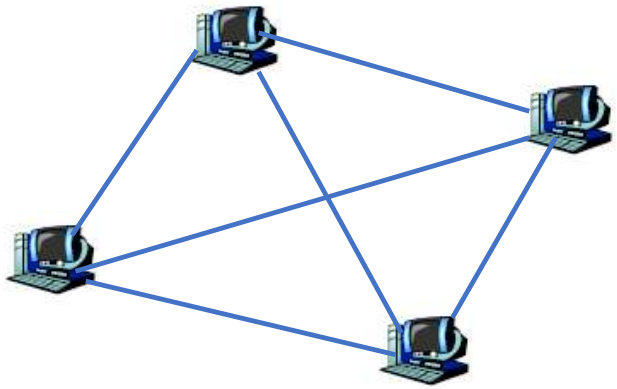
- **Pros:**
  - Scalable and easy to extend.
  - Supports hierarchical organization.
  - Failures may remain confined to a branch.
- **Cons:**
  - Higher-level nodes/links may become bottlenecks or single points of failure
  - Failure of an upper-level node can affect an entire subtree.

Introduction

Network Topologies: Mesh

- In the **Mesh** topology nodes are point-to-point linked in a non-hierarchical way:
  - **Full mesh**: all nodes connected (fully connected).
  - **Partial mesh**: nodes connected with some others.

- **Pros:**
  - Multiple paths between nodes → high robustness and fault tolerance.
  - Dedicated point-to-point links can reduce contention.
  - No single central point of failure.
- **Cons:**
  - Poor scalability in a full mesh.
  - High cost and complexity as the number of nodes grows.
  - Each node may require many interfaces/ports.

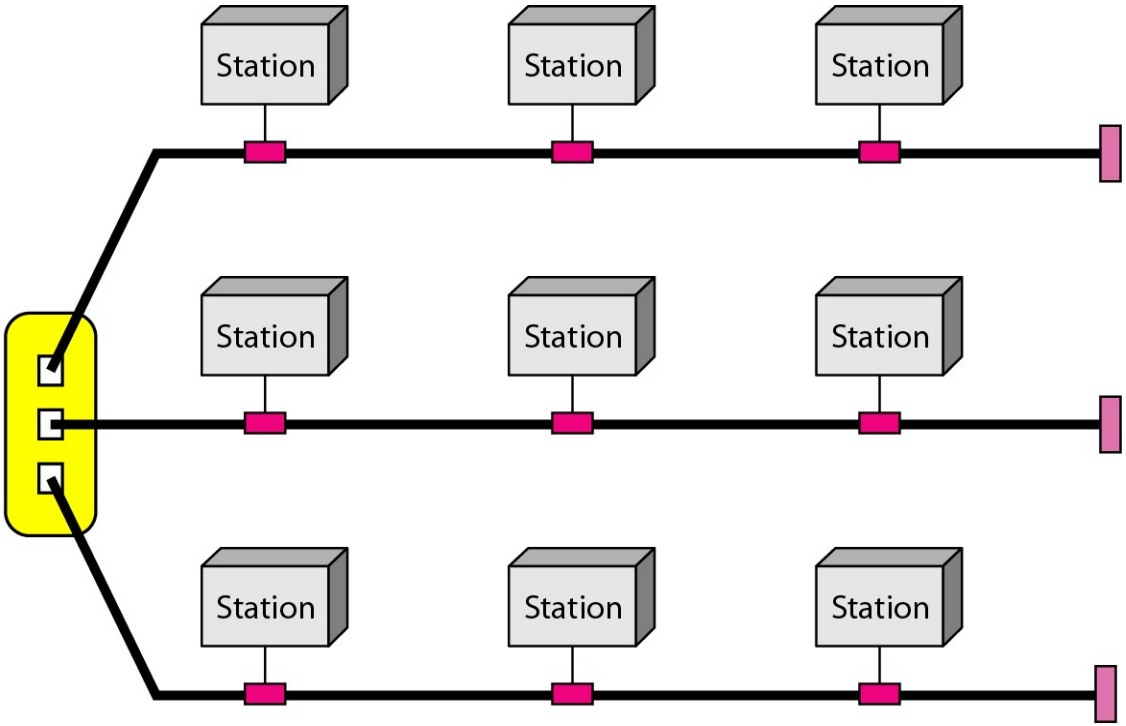


Full mesh networks have $n(n-1)/2$ links

Introduction

Network Topologies: Hybrid

- Topologies can be **mixed** into a hybrid network.

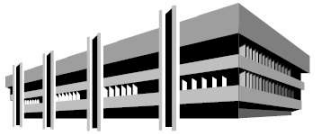


Star-backbone with 3 bus networks

Introduction

Categories of Networks

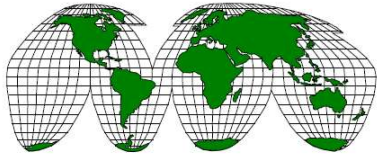
- Networks are categorized by size, number of hosts, bandwidth.
  - **Personal Area Network** (PAN)
  - **Local Area Network** (LAN) or Wireless Local Area Network (WLAN).
  - **Metropolitan Area Network** (MAN).
  - **Wide Area Network** (WAN)



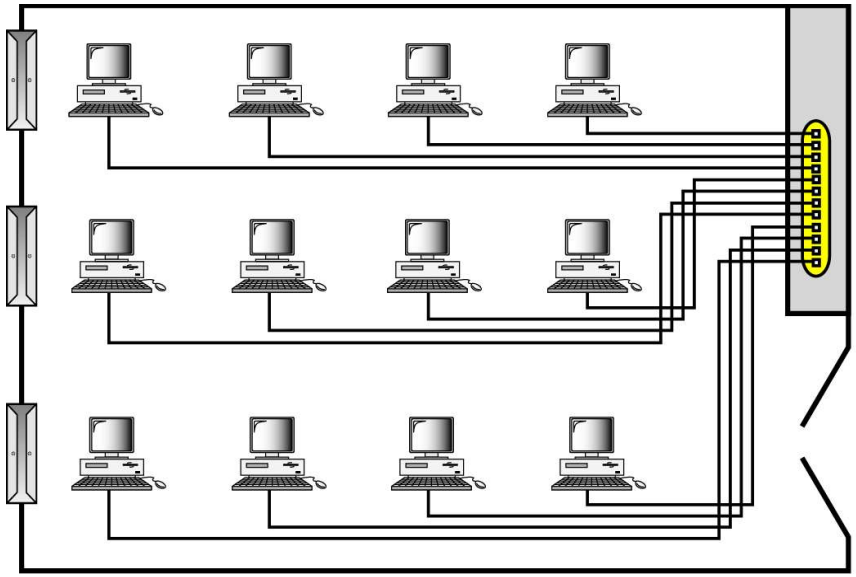
LAN/WLAN



MAN



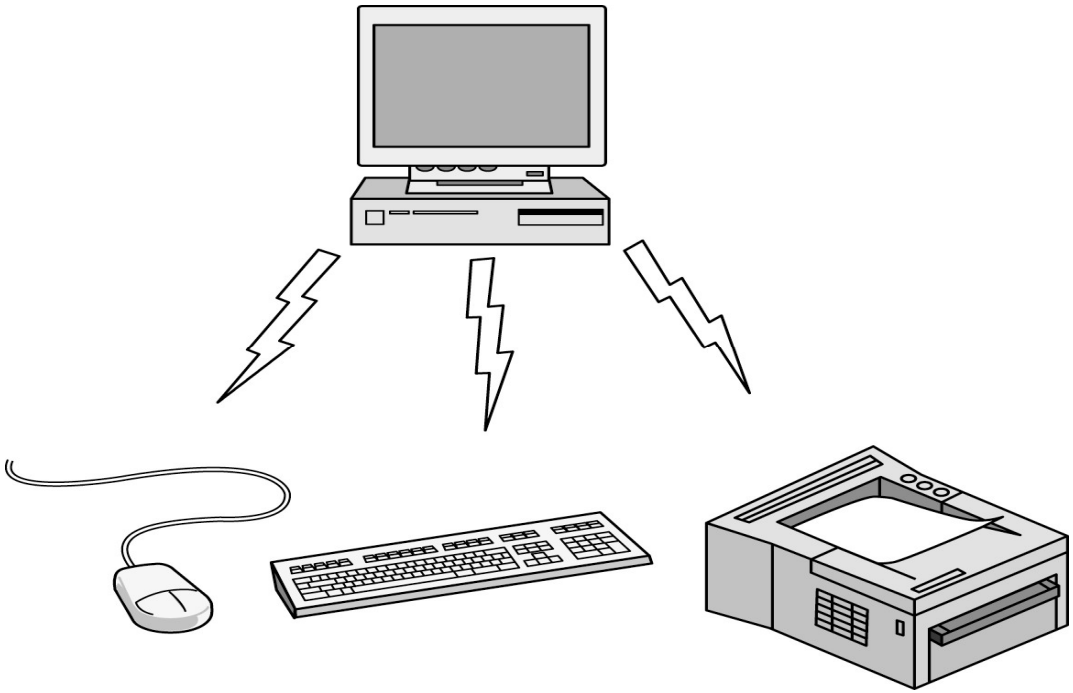
WAN



Example of LAN connecting 12 computers to a switch

Personal Area Network

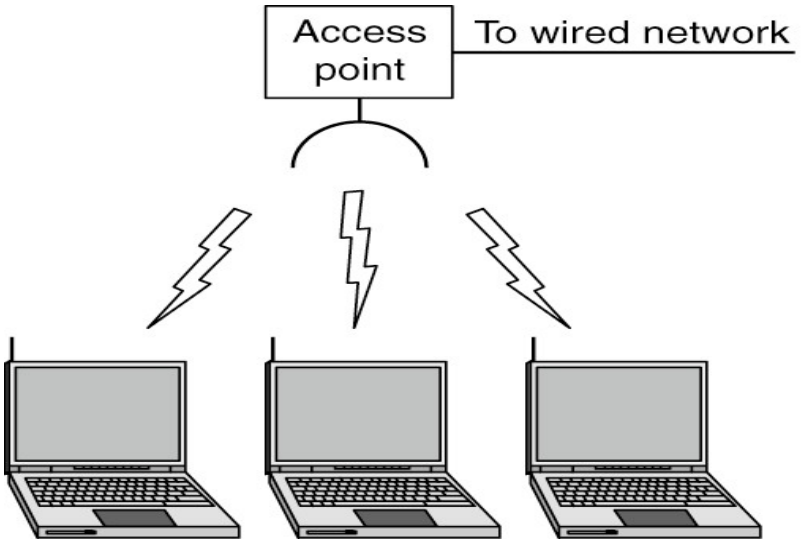
Categories of Networks

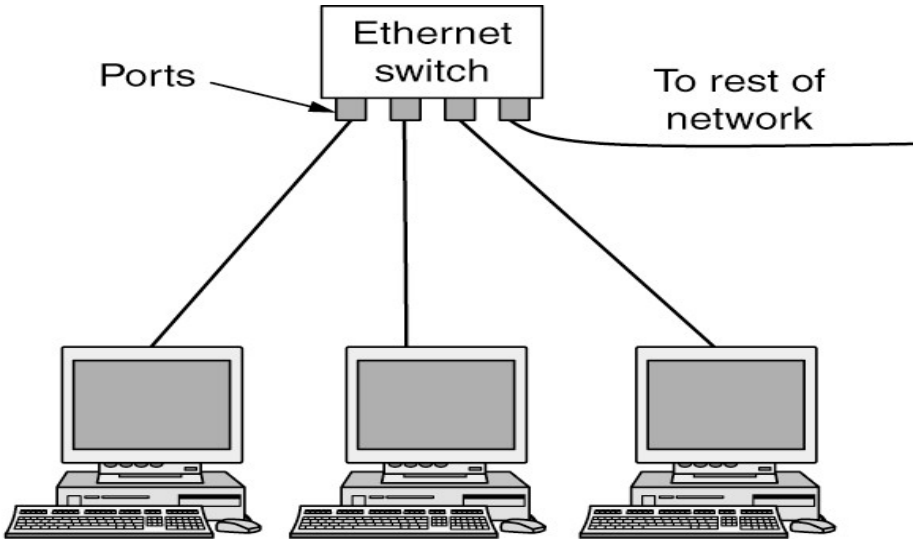


PANs (Personal Area Networks) let devices communicate over the range of a person. Bluetooth is a short-range wireless network used to connect components without wires.

Local Area Network

Categories of Networks





Left: IEEE 802.11 Wireless LAN (WLAN).

Right: wired switched Ethernet network.

Home Network

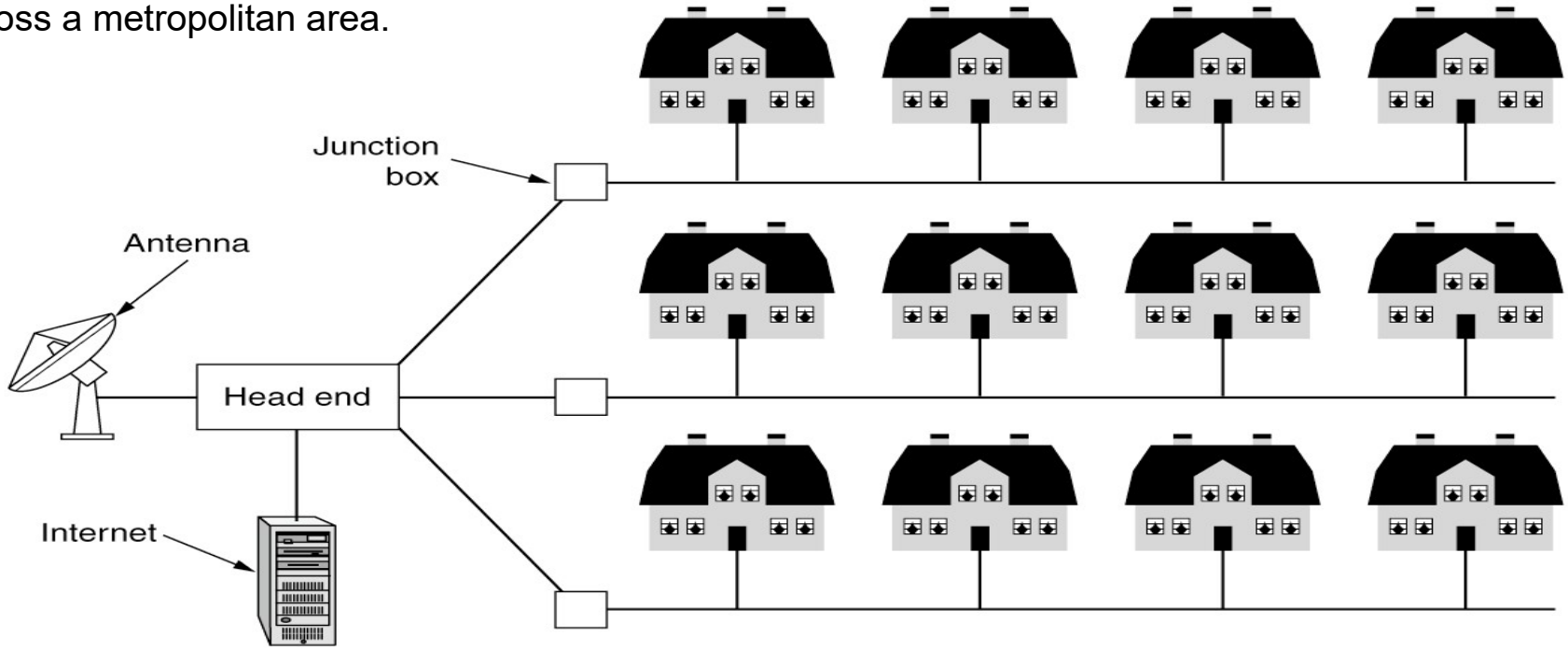
Categories of Networks

- Home network LAN
  - Broad, diverse range of Internet-connected devices
  - Characteristics: manageable, dependable, and secure
- Internet of things
  - Allows almost any device to connect
- Required home network properties
  - Easy to install
  - Secure and reliable
  - Interfaces work between all products
  - Reduced consumer device costs

Metropolitan Area Network

Categories of Networks

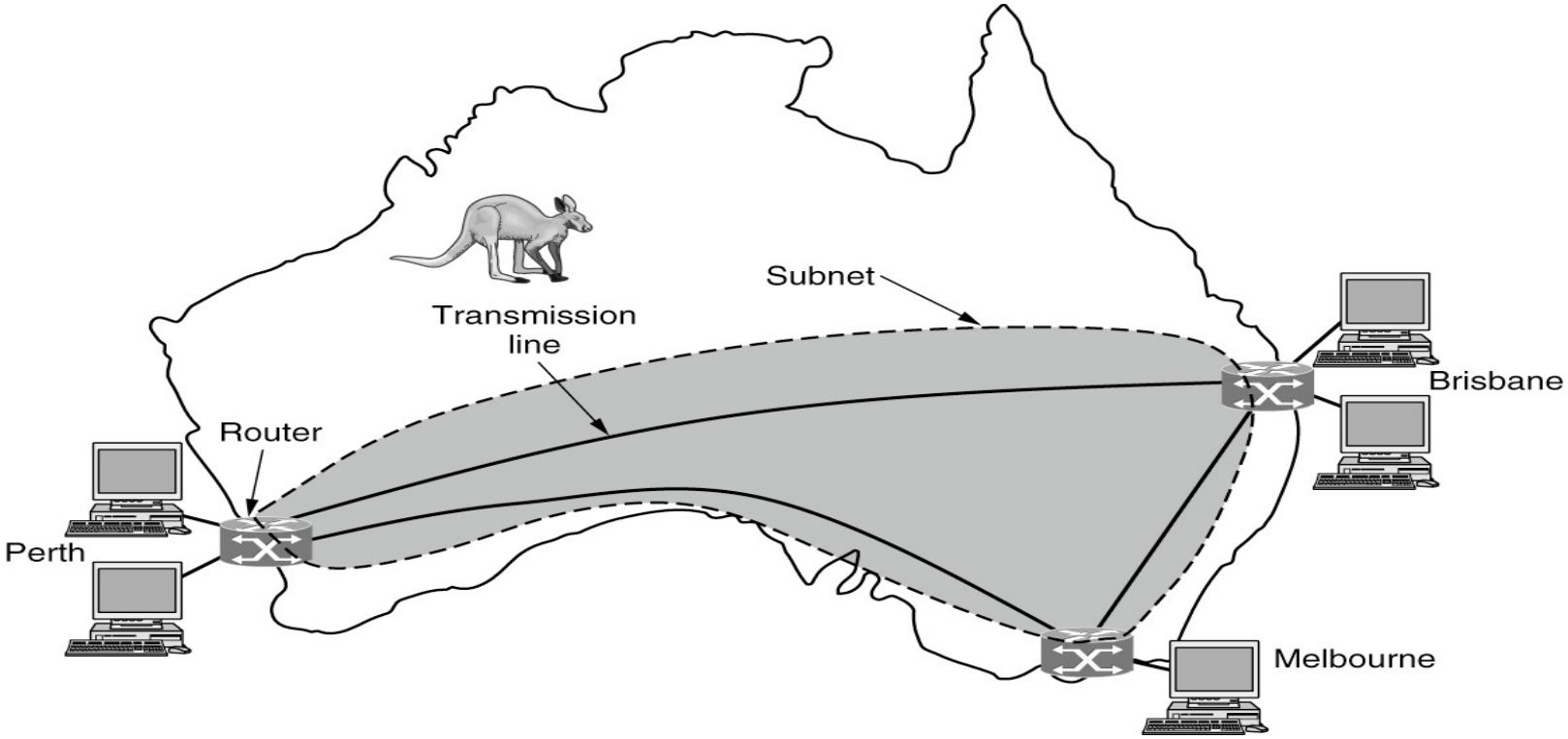
MAN = interconnection/aggregation of local networks or users across a metropolitan area.



A MAN (metropolitan area network) where both television signals and the Internet are being fed into the centralized cable head-end (or cable modem termination system) for subsequent distribution to people's homes.

Wide Area Network (1/3)

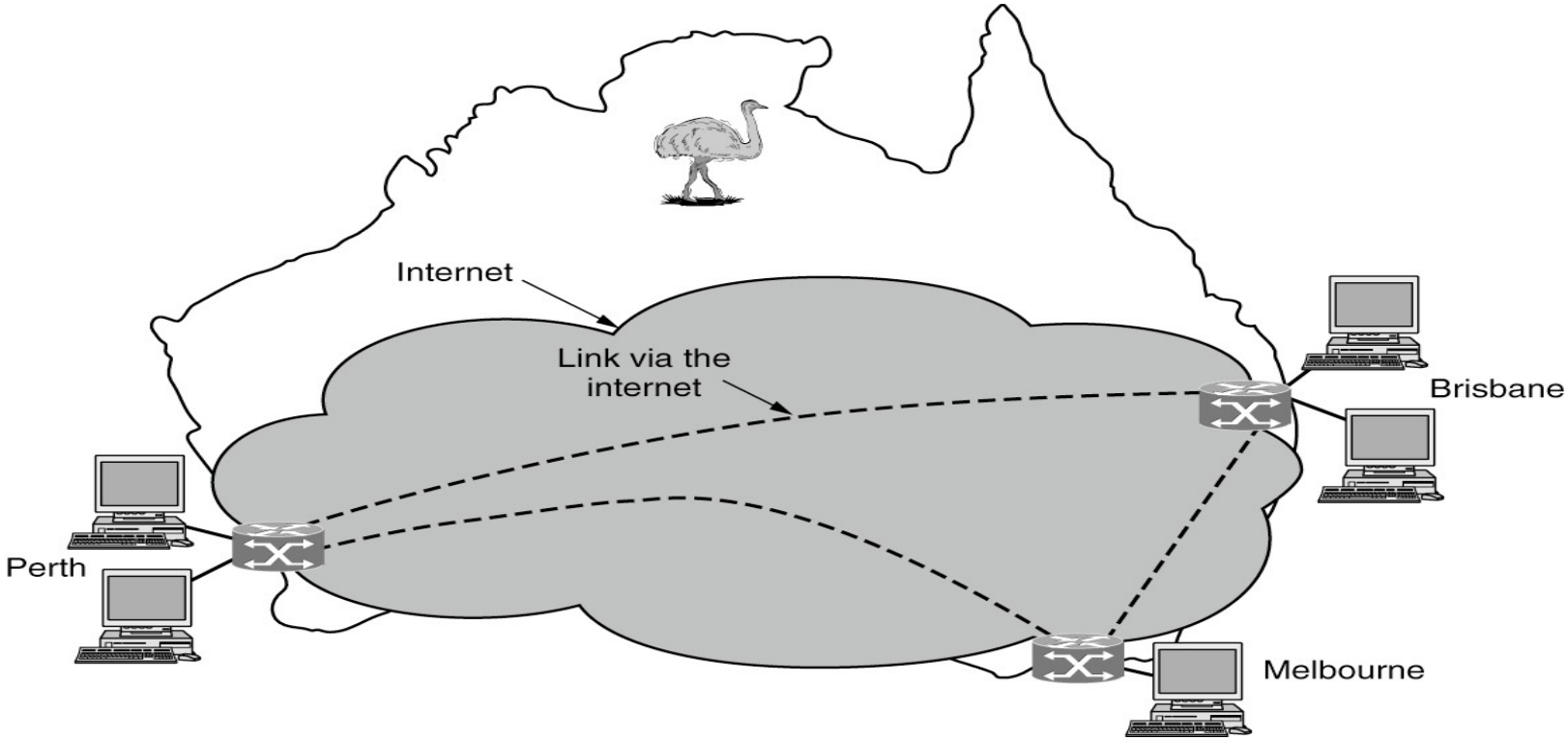
Categories of Networks



This wide area network illustrates how hosts in Perth, Brisbane, and Melbourne can communicate using leased lines.

Wide Area Network (2/3)

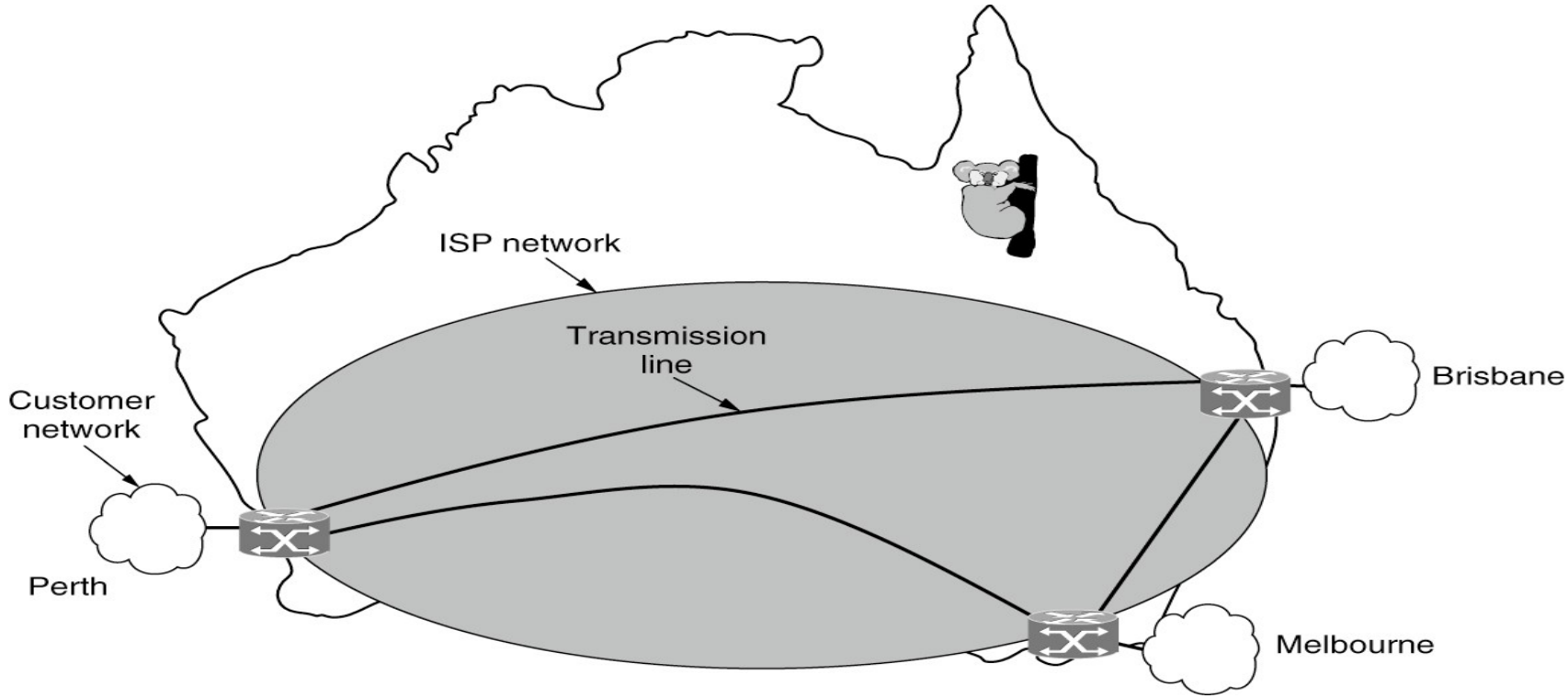
Categories of Networks



This wide area network illustrates how hosts in Perth, Brisbane, and Melbourne can communicate via the Internet.

Wide Area Network (3/3)

Categories of Networks

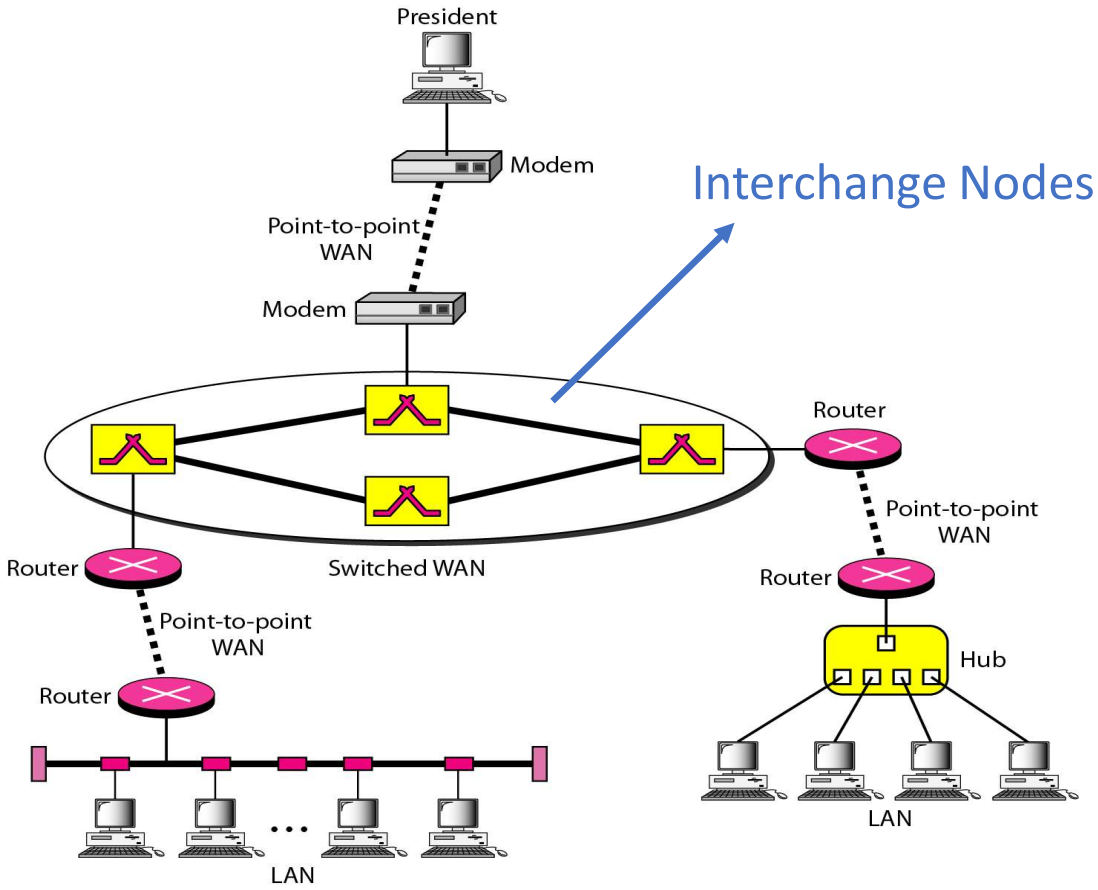


The sites are interconnected through the provider's private backbone.

Introduction

Complexity of WANs

- The complexity of the topology may increase with the size of the network.
- As the network grows, multiple intermediate nodes and paths may be required.
- WANs connecting nations or continents can obviously be very complex and heterogeneous.
- WANs may interconnect multiple LANs and point-to-point links through heterogeneous network technologies.



Example of heterogeneous WAN

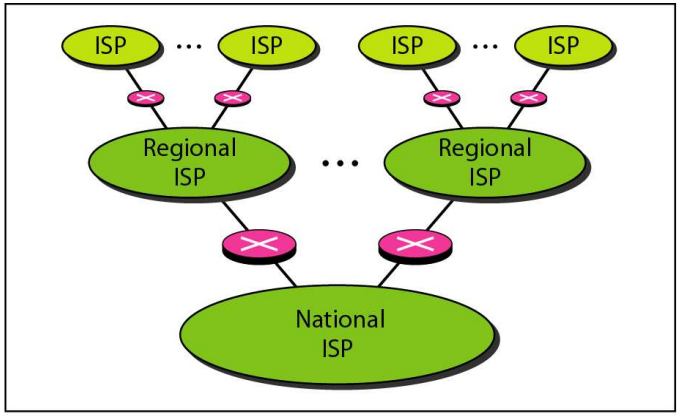
Introduction
Service Providers

- Computer networks enable access to **remote resources and services**.
- A network **resource** is a remote "object" or capability we want to have access: web pages, storage, computation, video or audio files, ...
- A network **service** is a functionality provided by a remote system: web access, e-mail, file storage, name resolution, remote computation, …
- The distinction between resources and services is not always sharp (e.g., let me *see* a *video*, *give* me access to a *web page*, are actions involving objects).
- One fundamental network service is Internet access, provided by **Internet Service Providers (ISPs)**.

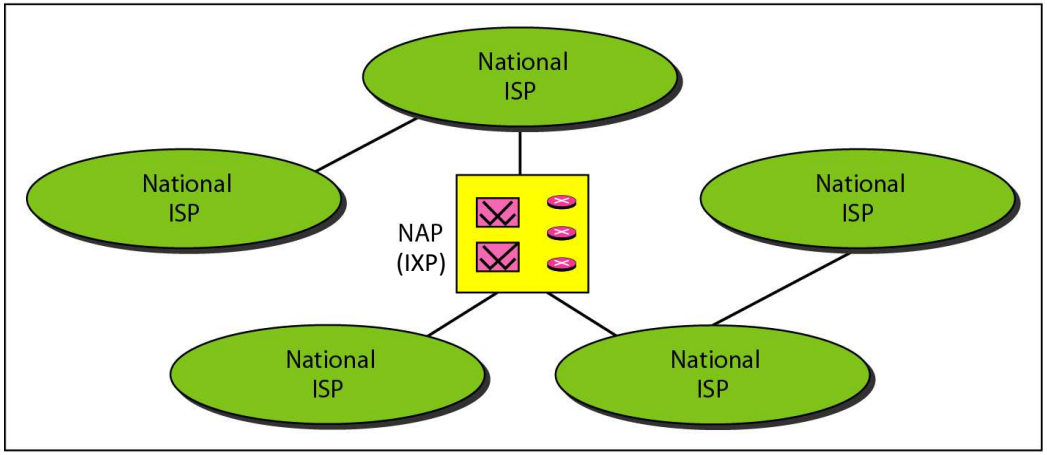
Introduction

Internet Service Providers

- **Internet access** is a fundamental service.
- An **Internet Service Provider** (ISP) provides Internet connectivity to users and organizations.
- ISPs may operate at different geographic scales and interconnect with other networks through **transit** and **peering**.
- ISPs can be organized as commercial, community-owned, non-profit, or privately owned
- Different ISPs exchange data through **Internet Exchange Points** (IXPs)
  - Old name **Network Access Points** (NAPs)



a. Structure of a national ISP

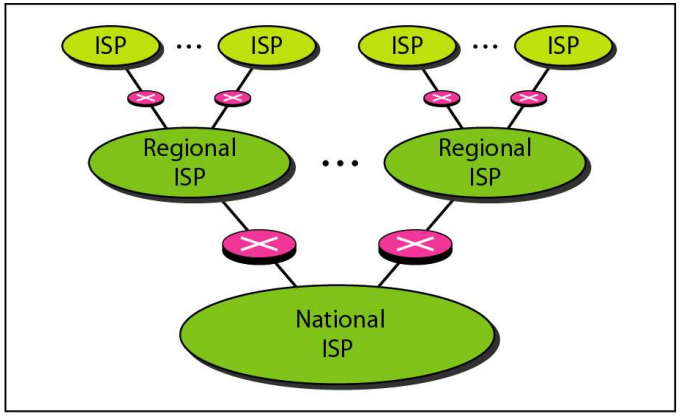


b. Interconnection of national ISPs

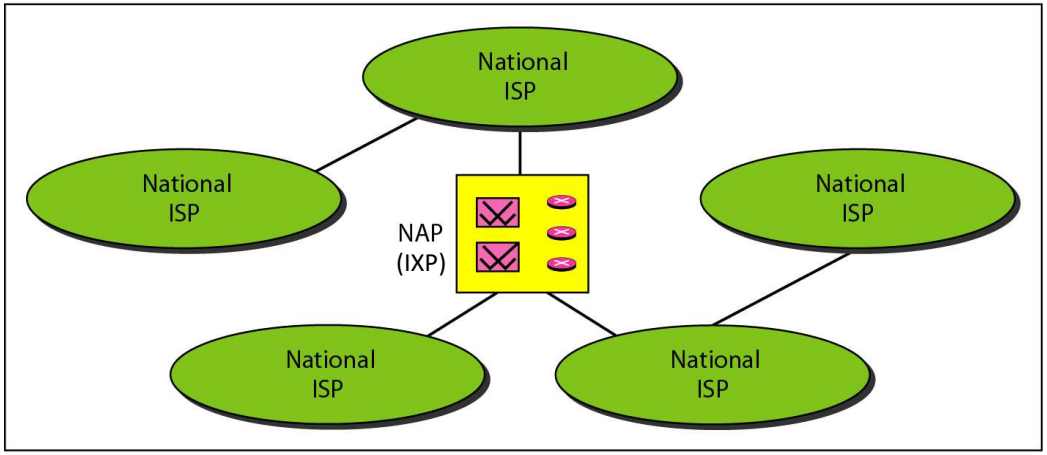
Introduction

Internet Service Providers

- ISP networks are typically organized around multiple PoPs interconnected by a backbone:
  - **Point of presence** (PoP): physical location with routers used by ISPs to connect customers or networks.
  - ISPs may operate at different geographic scales: access, regional, national, global.
- **Internet exchange point** (IXP) provide shared infrastructure where independent networks interconnect and exchange traffic.



a. Structure of a national ISP



b. Interconnection of national ISPs

Introduction

Example of the GARR Network

- GARR provides high-performance network connectivity and services to the Italian research and education community.
- GARR (Gruppo per l'Armonizzazione delle Reti della Ricerca) is the Italian national computer network for universities and research.
- GARR is interconnected with the European GÉANT network and, through it, with research and education networks worldwide.

